Breakout Factor Analysis

1 Objective

The objective of this test was to examine whether a stock's position relative to its previous 20-day closing high can predict its subsequent 20-day return.

The initial hypothesis was that stocks experiencing stronger breakouts might continue to outperform over the following 20 trading days.

2 Breakout Signal Construction

For each stock, the previous 20-day closing high was calculated while excluding the current day's closing price:

$$H_{20,t} = \max(C_{t-20}, \ldots, C_{t-1})$$

The breakout signal was defined as

$$Breakout_t = \frac{C_t}{H_{20,t}} - 1.$$

The subsequent 20-day return was calculated as

$$R_{t,t+20} = \frac{C_{t+20}}{C_t} - 1.$$

A positive breakout value indicates that the current close is above the previous 20-day high, while a negative value indicates that the stock remains below its previous high.

3 Breakout Signal Statistics

The observed distribution was:

Statistic	Value
Mean	-0.041861
Median	-0.030465
Standard deviation	0.051515
Minimum	-0.588026
25th percentile	-0.062847
75th percentile	-0.009006
Maximum	0.255618

The negative mean and median indicate that most observations were below the previous 20-day closing high.

4 Decile Analysis

Stocks were sorted into ten cross-sectional groups each day according to their breakout signal. Group 1 represents stocks furthest below their previous 20-day high, while Group 10 represents stocks closest to or above their previous 20-day high.

Group	Average 20-day Return
G1	1.9309%
G2	1.9448%
G3	1.9198%
G4	1.9726%
G5	1.8099%
G6	1.8134%
G7	1.6865%
G8	1.3931%
G9	1.2468%
G10	1.0997%

The low-minus-high spread was

$$1.9309\% - 1.0997\% = 0.8311\%.$$

The decreasing pattern suggests a reversal-type relationship rather than breakout continuation.

5 Information Coefficient

The daily cross-sectional Spearman Information Coefficient was calculated between the breakout signal and subsequent 20-day returns.

Metric	Value
Mean IC	-0.045372
Median IC	-0.034348
IC standard deviation	0.252073
Positive IC fraction	44.25%
t-statistic	-6.839809
p-value	< 0.001

The negative mean IC indicates that higher breakout values were associated with lower subsequent 20-day returns.

6 Long–Short Portfolio

A long–short portfolio was constructed by going long the lowest breakout decile (G1) and short the highest breakout decile (G10).

Metric	Value
Mean daily return	0.008311
Median daily return	0.005809
Daily standard deviation	0.069630
Annualized return	209.4489%
Annualized volatility	110.5338%
Sharpe-like ratio	1.8949
Ordinary t-statistic	4.535928
Ordinary p-value	0.000006

The ordinary test indicates a statistically significant positive long–short spread.

However, the 20-day forward returns overlap across consecutive observations, which creates serial dependence.

7 HAC / Newey–West Test

To account for the overlapping 20-day forward returns, a Newey–West HAC standard error with 20 lags was used.

Metric	Value
Mean return	0.008311
HAC standard error	0.004635
HAC t-statistic	1.793215
HAC p-value	0.072939

The HAC p-value is

$$0.072939 > 0.05.$$

Therefore, the long–short return is not statistically significant at the 5% level after accounting for overlapping returns.

8 Limitations

The primary statistical issue is the overlap in 20-day forward returns. Consecutive observations share many of the same future trading days, creating serial dependence. The HAC/Newey–West test was therefore used for the final statistical inference.

The annualized return of 209.4489% should also not be interpreted as an expected annual investment return. It results from mechanically annualizing the mean return obtained from overlapping 20-day forward returns.

9 Conclusion

The Breakout analysis provides evidence of a negative relationship between the breakout signal and subsequent 20-day returns.

The decile analysis produced a low-minus-high spread of

$$\boxed{0.8311\%}$$

and the mean IC was

$$\boxed{-0.045372}.$$

These results indicate a reversal-like relationship: stocks farther below their previous 20-day high tended to have higher subsequent 20-day returns than stocks near or above their previous high.

However, after correcting for overlapping observations, the HAC test produced

$$t_{\mathrm{HAC}} = 1.793215, \qquad p_{\mathrm{HAC}} = 0.072939.$$

Since the HAC p-value is greater than 0.05, the Breakout factor should not be treated as a statistically confirmed standalone alpha factor at the 5% significance level.

Instead, the evidence suggests that Breakout may be useful as a supporting reversal-related signal in a later multi-factor model.

$$\boxed{\text{Breakout: Supporting / Potential Reversal Factor}}$$